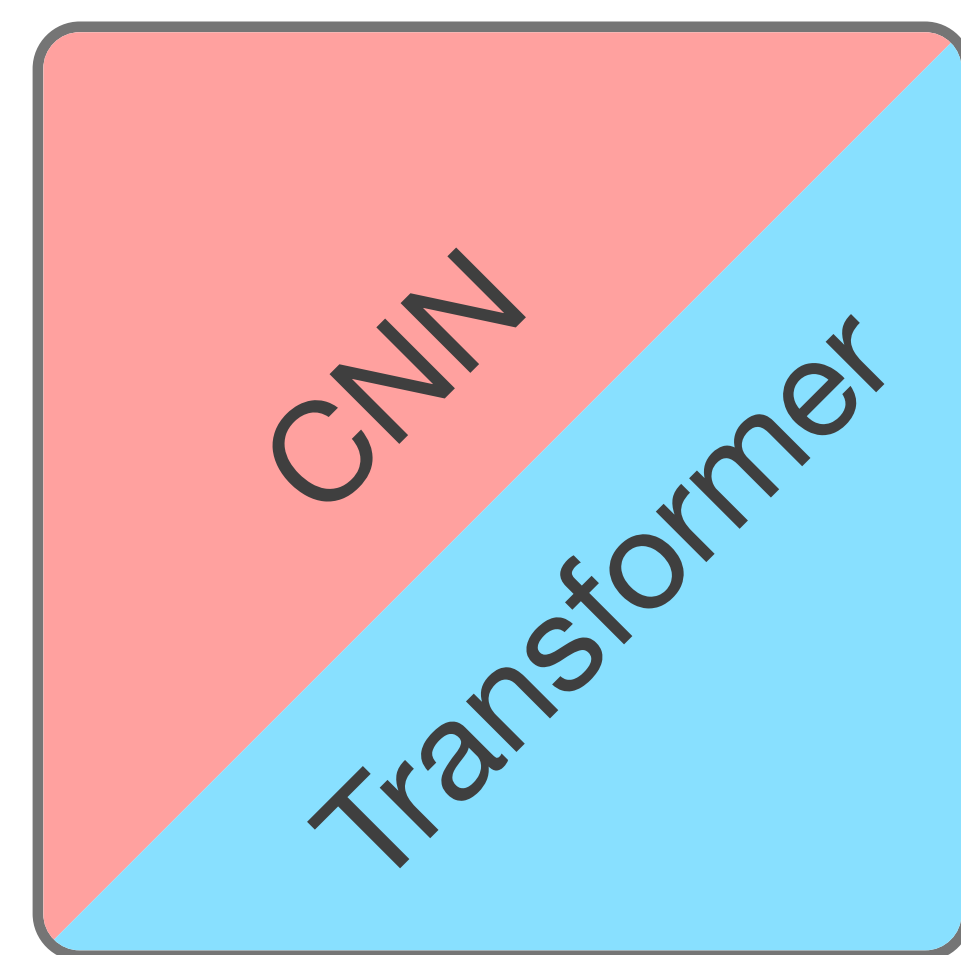
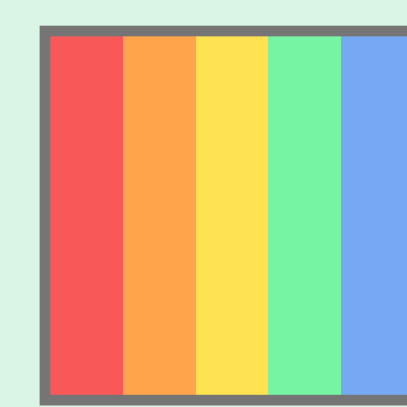
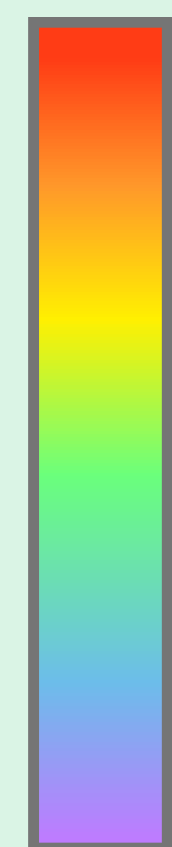




Visual Embedding

Classifier Weight

f

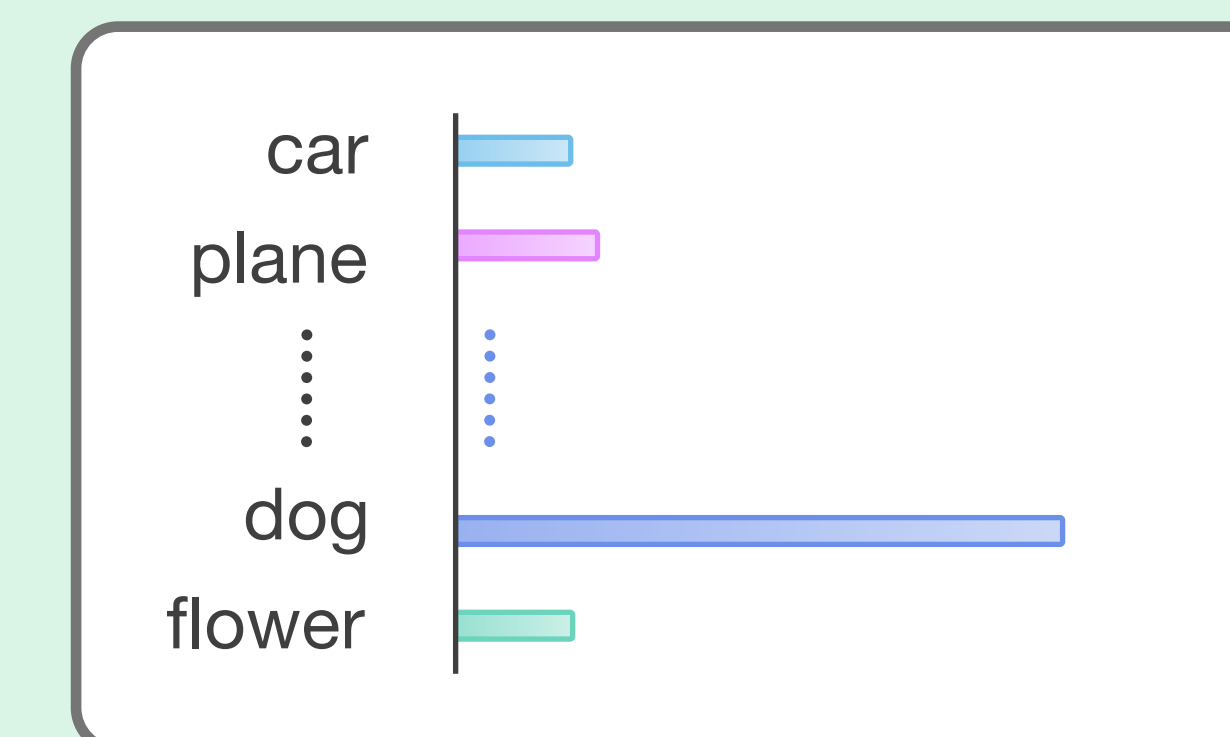


W



MatMul

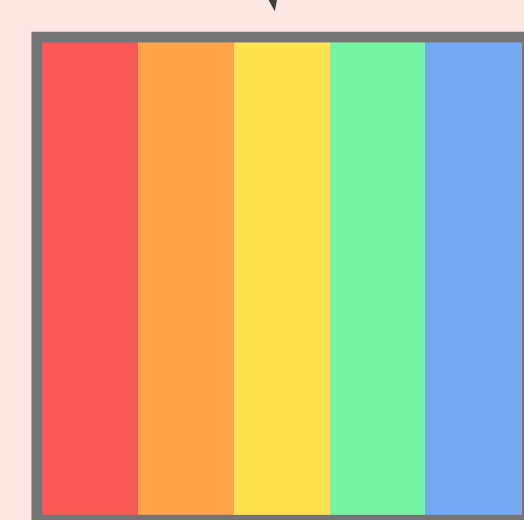
Naive Classification



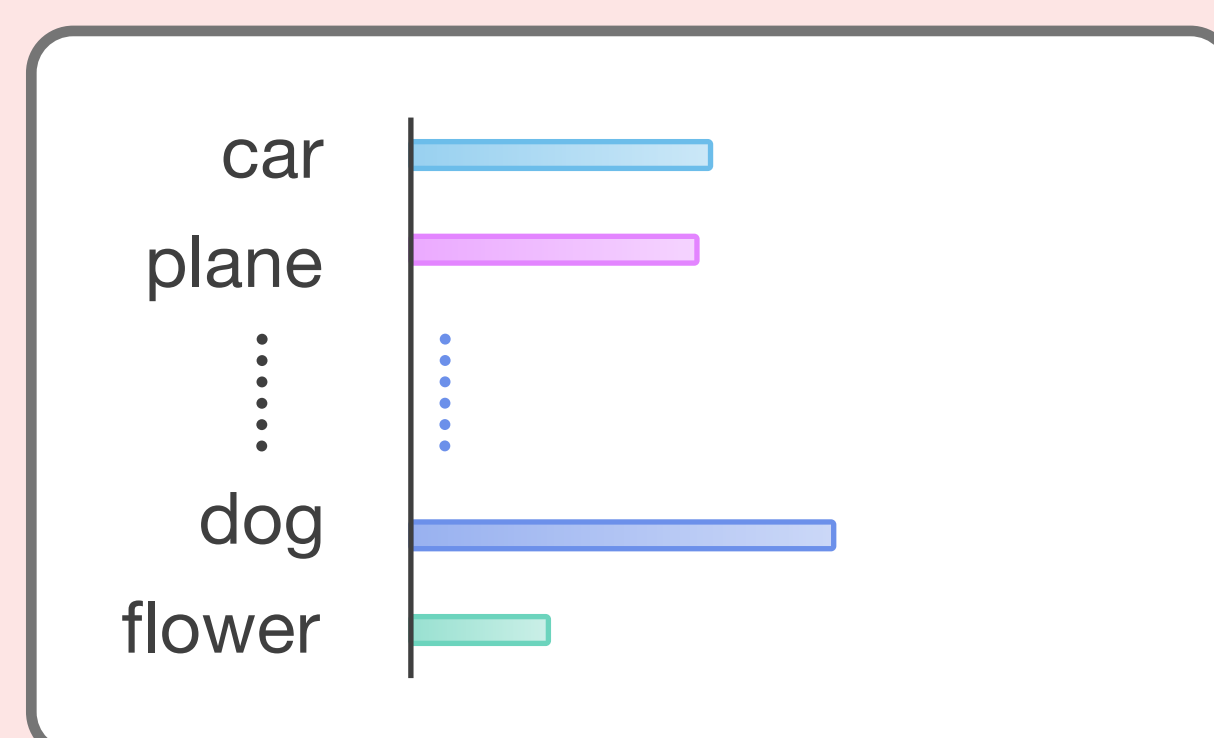
car plane dog flower

[CLS] a photo of a dog

Text-Encoder



W

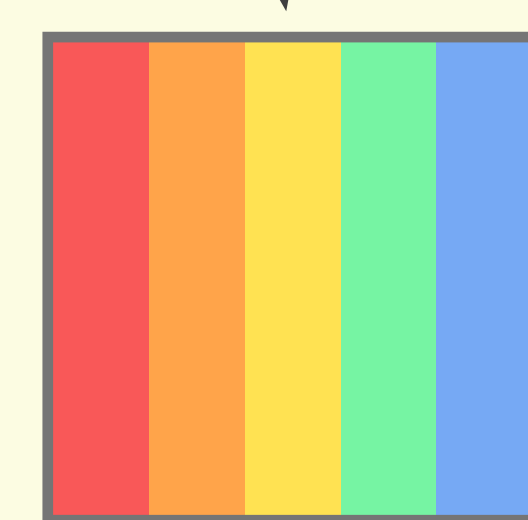
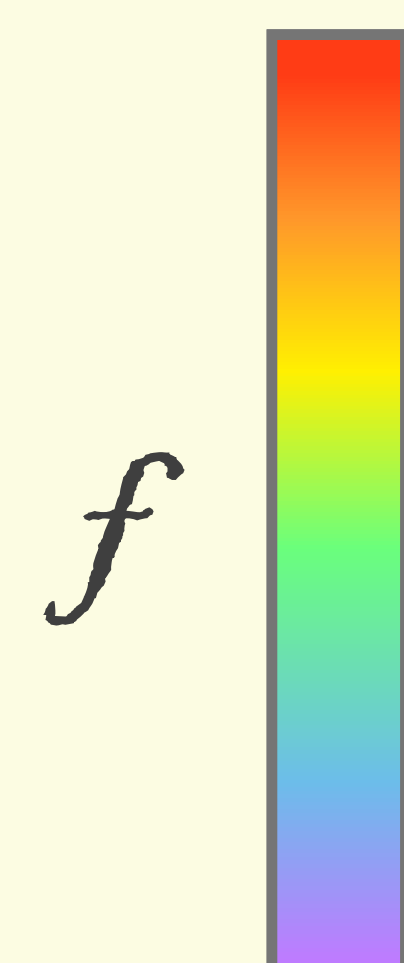


CLIP

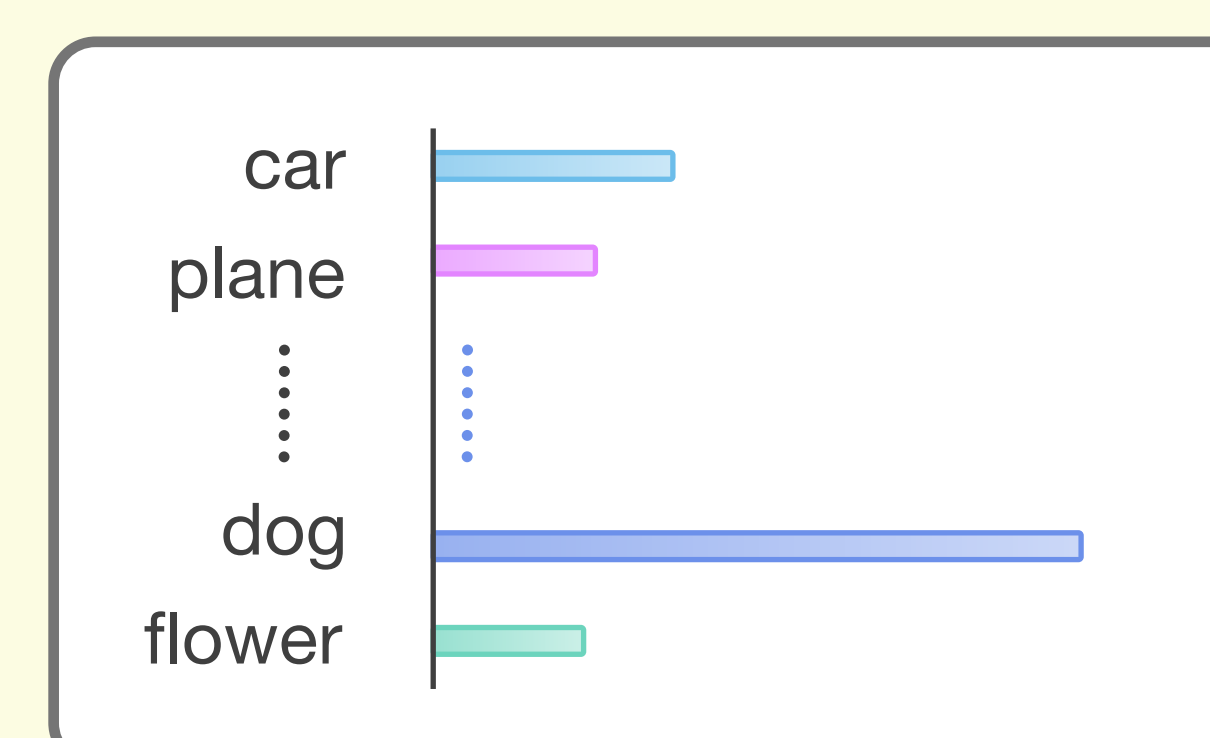
car plane dog flower

[CLS] [V]₁ [V]₂ [V]_n dog

Text-Encoder



W



CoOp

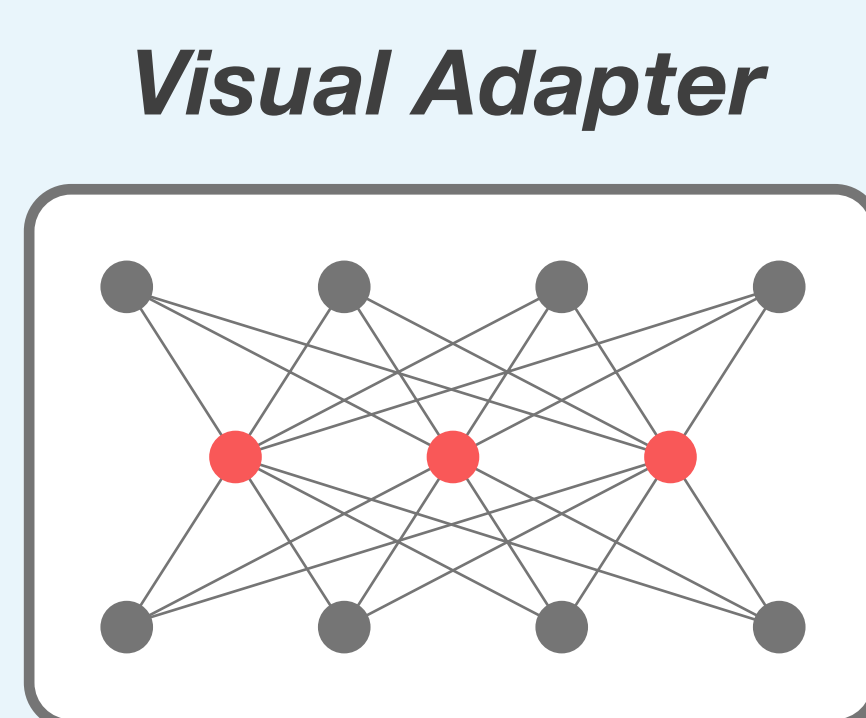
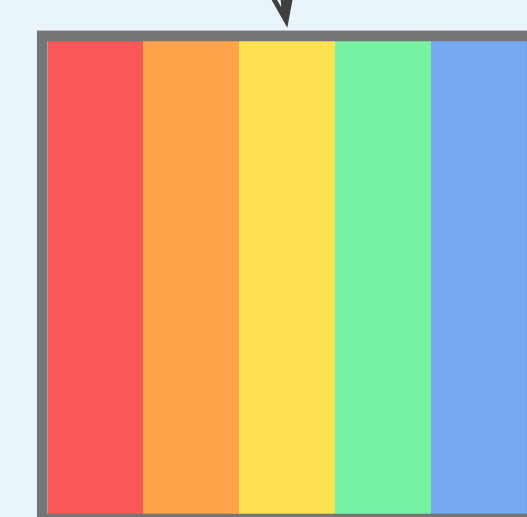


car plane dog flower

[CLS] a photo of a dog

Text-Encoder

W



Visual Adapter

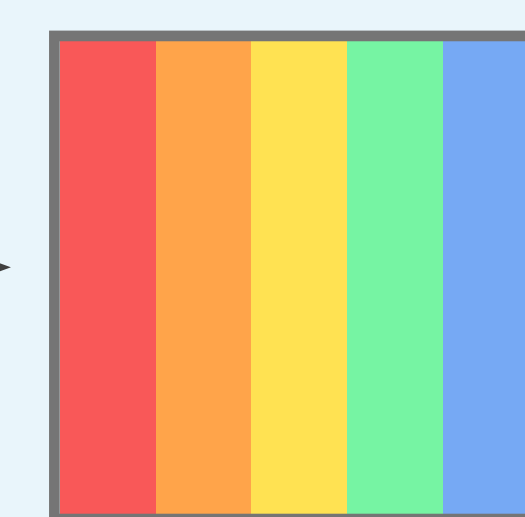
$Av(f)$

+

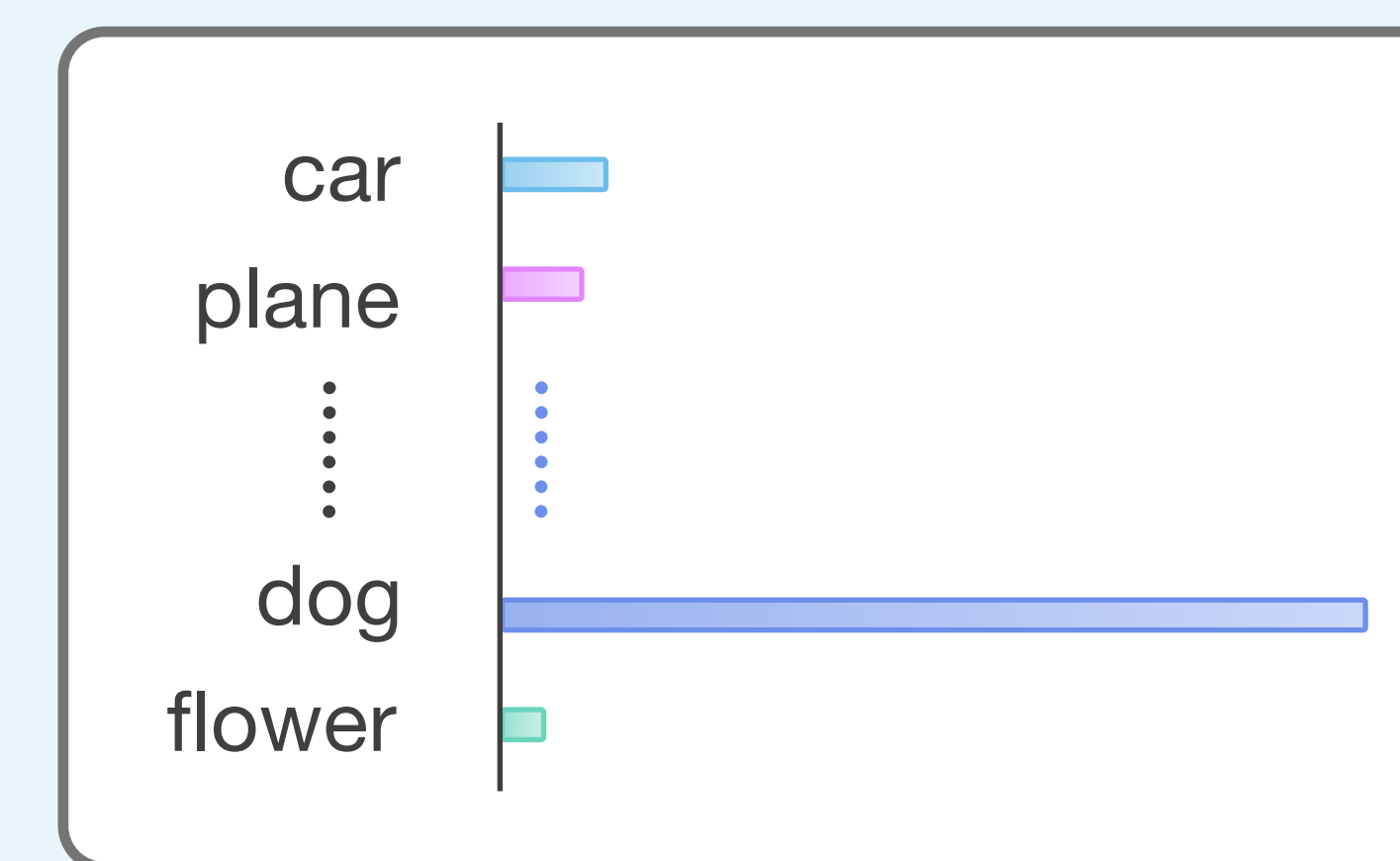


$At(W)$

+



+



CLIP-Adapter